

The move is completed with the microinstruction fields:

- °°FF1 $\overline{FO} \leftarrow 1$
- °°AN $CI \leftarrow \overline{FO}$ for both arrays.
- °°REE Selects the T register of both arrays.
- °°KD Selects the Data Out of the arrays as input to the K-Bus.
- °°ED1 By default (ED2) selects Array 2 as source of Data Out Bus.
- °°LDI CPE function.

The functional equation for LDI is:

$$T \leftarrow (I \quad K) - 1 + CI.$$

Since CI to both arrays is a one, the last two terms cancel.

The I-Bus has been forced to all ones, which leaves the K-Bus unaltered. As the K-Bus is being driven by the DO Bus of the high-order array, the LDI function actually performed looks like:

$$T \text{ (8080 Acc)} \leftarrow DO_2 \text{ (containing the value of H).}$$

Thus, LDI completes the functional execution of MOV A,H. All that remains is to return the microprogram to its Fetch cycle (006₁₆). 07F₁₆ has already enabled the fetch signals:

- °°EXT Selects $\overline{ReadyQ}$ as source for MA₀.
- °°IRW Latches external Data Bus into Instruction Register.
- °°IST Enables interrupt acknowledge logic.

With the signals on the previous page already active for one microcycle, $\overline{Ready}$ may be returned immediately, and the microprogram may proceed directly to 006₁₆. If $\overline{Ready}$ is not returned immediately, the next microinstruction executed will be 007₁₆, and the microprogram will wait for the external memory to respond with the next op code.

A summary of the MOV A,H macro instructions is presented in Figure 15 in the form of a microcode listing.